MINING DEALER BEST PRACTICE

PLATE EAR CONTROL TOOL

Maintenance and Repair | Application | Component Life Management | **Component Renewal (CRC)** | MARC Management

April 2021
4.03-210430-1

1.0 Introduction

When considering the how reusable plates are in a transmission, the condition of the plate is inspected as is the dimensional conformance of the plate. The profile and dimensions of the tangs/ears on the transmission plates are inspected during rebuild to establish whether they can be re-used.

2.0 Best Practice Description

Each plate has 5 or 6 tangs/ears, each containing a slot. The slot dimensions vary according to the model. The conformance of each of the slots should be checked in accordance with salvage guideline SEBF8257.
The Plate Ear Control Tool is machined and acts as a go/no-go to quickly determine the suitability and conformity of the slots in the clutch plates. Slots shall be inspected using a tool based on the maximum size.
This go/no-go tool saves a lot of inspection and recording time (For example, a transmission of D8T has 2 pistons, 13 plates with 5 slots in each plate. That is 90 measurements and these measurements also have records). The simple drawing below showing the tool can be used in all transmissions with pin diameter 12.64mm (Plate numbers include 8P-2051, 6I-8500, 7G-0437, 232-7432, 385-8958). Some pistons also have slots in this way so can also utilize this tool (9G-0280 & 9G-0285).

Wear Limits for the Slot for the Pin		
Diameter of the Reaction Pin	**Maximum Width of the Radial Distortion Slot (P)**	**Maximum Width of the Slot (R) for Either Direction**
12.7 mm (0.500 inch)	16.26 mm (0.640 inch)	7.76 mm (0.31 inch)

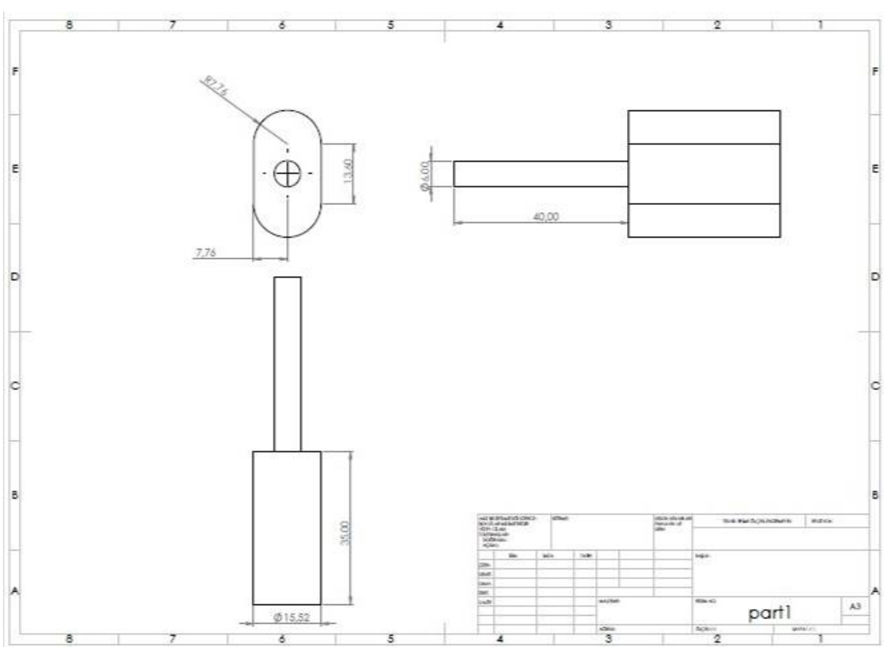


Plate Ear Control Tool	DATE 04/30/2021	CHG NO 00	NUMBER 4.03-210430-1

3.0 Implementation Steps

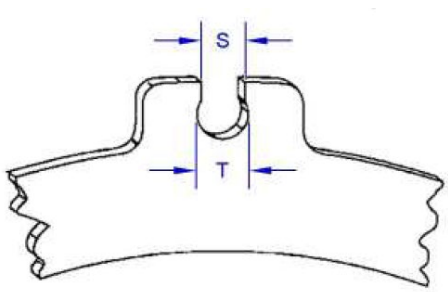
Step 1: Flat surfaces on tool are kept towards to technician.

Step 2: Usability on both sides of the tool is checked to ensure that the radiused portions adjusted to the maximum size fit into the ovalized areas on the slots

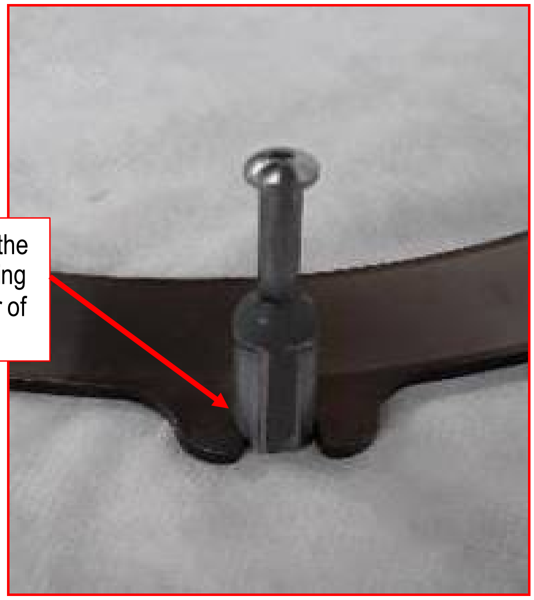
Step 3:

- Tool enters the slot => Re-use
- Tool does not re-enter the slot => Do not re-use
- Tool has excessive clearance in the slot => Do not re-use

Reusability status is checked by determining whether the apparatus passes through the Slot holes in the Plate ears.

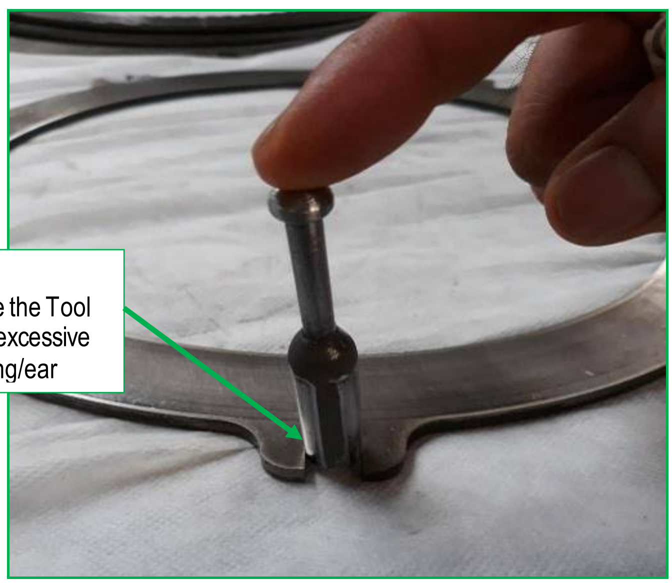


Part Number	Number of Tangs (B)	Width of Tang (M)	Width of the Slot for Pin (L)	Maximum Width of Radial Distortion	Maximum Width of the Slot	Other Identifying Features
8P-2051	5	34.0 mm (1.34 inch)	12.98 mm (0.511 inch)	7.76 mm (0.306 inch)	16.256 mm (0.6400 inch)	A relief for tang exists.



Gap alongside the Tool representing excessive wear of slot in tang/ear

Not Reusable



- Tool fits in slot
- No gap alongside the Tool representing no excessive wear of slot in tang/ear

Reusable

4.0 Benefits

Reusability status is checked by determining whether the tool's passes through the Slot holes in the Plate ears.

1) Plate slot inspection time is reduced.
2) The go/no-go tool decreases risk because each measurement has a margin of error, with the risk reduced when the same work is done with less measurement.

5.0 Resources Required

Labor :3 hours

6.0 Supporting Attachments / References

None.

7.0 Related Best Practices

None.

8.0 Acknowledgements

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